SURJECTIVITY ALONE DETECTS FINITE PRESENTATION IN THE LIMIT CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 817-852, following the statement at 801-815.

Editorial note. The parenthetical 'modulo a set theoretic remark exactly as in the proof of Proposition ...' is a by-reference bookkeeping note, flagged.

Allowed standard prerequisites (editorial).

- Scheme-level surjection criterion. Limits, Lemma limits-lemma-surjection-is-enough (cross-chapter, cited at ~825-826). U -> V is lfp from the surjectivity alone.
- Characterization of locally finite presentation spaces. proposition-characterize-locally-finite-presentation (same chapter, card L26289, cited at ~811-814 and ~848). Surjectivity for the etale cover h : U -> X x_Y V (h is lfp).
- Relative limit preserving criterion. lemma-characterize-relative-limit-preserving (same chapter, cited at ~813-814). Part (2) of the proposition whose surjectivity-only form is proved.

Source reference. Original lemma 06BC: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 351–351. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CM6: focused-seeds/geometry/S9-spaces-limits-surjection-is-enough/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If for every directed limit $T = \lim_{i \in I} T_i$ of affine schemes over S the map*

$$\operatorname{colim} X(T_i) \longrightarrow X(T) \times_{Y(T)} \operatorname{colim} Y(T_i)$$

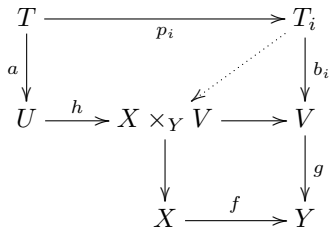
is surjective, then f is locally of finite presentation. In other words, in Proposition 04AK part (2) it suffices to check surjectivity in the criterion of Lemma 06BC.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a scheme V and a surjective étale morphism $g : V \to Y$. Next, choose a scheme U and a surjective étale morphism $h : U \to V \times_Y X$. It suffices to show for $T = \lim T_i$ as in the lemma that the map

$$\operatorname{colim} U(T_i) \longrightarrow U(T) \times_{V(T)} \operatorname{colim} V(T_i)$$

is surjective, because then $U \to V$ will be locally of finite presentation by Limits, Lemma 0CM6 (modulo a set theoretic remark exactly as in the proof of Proposition 04AK). Thus we take $a : T \to U$ and $b_i : T_i \to V$ which determine the same morphism $T \to V$. Picture



By the assumption of the lemma after increasing i we can find a morphism $c_i : T_i \to X$ such that $h \circ a = (b_i, c_i) \circ p_i : T_i \to V \times_Y X$ and such that $f \circ c_i = g \circ b_i$. Since h is an étale morphism of algebraic spaces (and hence locally of finite presentation), we have the surjectivity of

$$\operatorname{colim} U(T_i) \longrightarrow U(T) \times_{(X \times_Y V)(T)} \operatorname{colim}(X \times_Y V)(T_i)$$

by Proposition 04AK. Hence after increasing i again we can find the desired morphism $a_i : T_i \to U$ with $a = a_i \circ p_i$ and $b_i = (U \to V) \circ a_i$. □